

Toward an Open Physics/AI Framework for the Crystallographic Phase Problem

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Working draft / software package v0.10.0 (MIT)

Code & data: https://github.com/pileofflapjacks1/grok_phase_solver

PyPI: <https://pypi.org/project/grok-phase-solver/>

Release tag: v0.10.0

Reviewer one-pager: [FOR_REVIEWERS.md](#)

Figures: [figures/paper_figure_captions.md](#)

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Status: Not submitted. Numbers frozen to repo scoreboards under [data/processed/](#).

Abstract

We present *grok_phase_solver*, an open Python framework that unifies classical solutions of the X-ray crystallographic phase problem—charge flipping, hybrid input-output (HIO), relaxed averaged alternating reflections (RAAR), difference-map projections, Patterson and direct methods, isomorphous difference Patterson, and density modification—with hybrid and learned phase priors (AI-PhaSeed, GraphPhaseNet, optional PhAI). Algorithms act on measured amplitudes $|F(hkl)|$ and are evaluated with origin-invariant map correlation (mapCC), peak recovery, R_1 , free figures of merit based on a positivity residual R_+ , and a strict multi-criterion success definition.

On easy synthetic cells, multistart free-FOM **ensemble** phasing is competitive with or better than local academic SHELXS under our scoring protocol. On hard cells ($n \gtrsim 12$, $d_{\min} \gtrsim 1.5$ Å), pure ab initio methods—including scaled graph priors—remain ~0% strict success. An **oracle partial- φ** experiment shows that when $\geq \sim 30\%$ of strong $|E|$ phases are correct within $\sim 20^\circ$, AI-PhaSeed extension strict-solves those hard cells, identifying **seed quality** as the hard-region bottleneck rather than free-FOM inversion. Scaling GraphPhaseNet through Wilson-matched and Melgalvis-style curricula (v3–v8, including d_in=22 multipath / HA-aware features and bin-classification losses) does **not** lift mean strong-phase accuracy past the **$\sim 30\%$** seed bar on laptop pilots (typical hold-out frac\$ $\$20^\circ \sim \mathbf{21\text{--}24\%}$; XL Melgalvis N=1200 $\approx \mathbf{22\%}$). On experimental COD Fobs, PhAI hybrids strict-solve COD 2016452 at 1.0 Å in our pipeline budget; a **fragment-half** path (SG-expanded partial model + full F_{calc} soft prior) reaches mapCC **comparable to or better than** oracle 30% partial- φ on two COD Fobs cells under the same short budget—while pure ab initio **auto** remains ~ 0.20 mapCC.

We ship scientist-facing tools (**gps-solve**, **gps-make-seed**, Streamlit **gps-gui**) exporting density maps and SHELXL-ready **trial.res**. We do **not** claim a general macromolecular ab initio solution or industrial equivalence to SHELXT on all cases.

1. Introduction

Recovering phases $\varphi(\mathbf{h})$ from amplitudes alone is the classical crystallographic phase problem:

$$\rho(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} |F(\mathbf{h})| e^{i\varphi(\mathbf{h})} e^{-2\pi i \mathbf{h} \cdot \mathbf{r}}.$$

Industrial small-molecule pipelines (SHELXT/SHELXS + SHELXL/Olex2) solve most atomic-resolution organics. Harder synthetic and experimental regimes, open science, and hybrid AI methods still benefit from transparent, modular baselines with **honest** failure reporting. Recent neural work (e.g. PhAI; Melgalvis–Rekis GraPhAI-style diffraction graphs) shows strong domain-specific results when packing and weights are carefully matched.

Contributions

1. **Integrated open stack** — classical solvers, free FOM, hybrid polish, learned priors, experimental I/O, optional SHELXS runners (binaries not redistributed), CLI and GUI.
2. **Hard-region science** — failure taxonomy; partial- φ seed bar (30% / 20°); negative scale result for pure GraphPhaseNet priors across v3–v8.
3. **Product path** — easy $\rightarrow$ ensemble; hard $\rightarrow$ partial- φ / **fragment** / **predicted-model** / HA seeds; `trial.res` $\rightarrow$ SHELXL.
4. **Calibration** — SHELXS H2H; experimental COD Fobs scoreboard; Wilson domain-gap matching; COD hard-path fragment validation.
5. **Realistic synthetics** — Melgalvis & Rekis (2026) style volume + artificial-molecule generation, multi-fragment packing, and HA-heavy curricula for prior training (`synthetic_melgalvis.py`).

2. Methods

2.1 Classical and projection algorithms

Implemented: charge flipping; HIO; RAAR; difference map; direct methods (E -values, triplets, tangent formula multi-start); Patterson peak picking; difference Patterson for heavy-atom vectors; Blow–Crick-style SIR/MIR FOMs; solvent flattening / density modification. An experimental **Hybrid Difference Map (HDM)** path blends DiffMap updates in ordered density with HIO feedback in solvent (`--method hdm`; research-only, not used by `auto`). Math notes: `docs/math/`.

2.2 Free figure of merit and ensemble

Truth-free ranking uses a composite free FOM whose amplitude residual is a **positivity residual** R_+ (not the vacuous post-modulus R of early free-FOM designs). Multistart **ensemble** (CF + RAAR) selects the best free-FOM trial and is our strongest *in-repo* ab initio path on easy cells (Fig. 2).

2.3 AI-PhaSeed and partial seeds

AI-PhaSeed (Carrozzini *et al.*, 2025; PhAI foundation Larsen *et al.*, 2024) selects strong- $|E|$ seeds from an AI phase vector, extends by density modification with seed re-imposition, and optionally polishes under free-FOM gate. The stack includes a **modified-tangent DM+AI hybrid** (AI phases as a priori weights), a **seed-quality Class 0/1 diagnostic** (heuristic features: max W , N_{asym} , Vol, seed fraction, free-FOM proxies, plus bin-entropy / $|\cos \varphi|$ cues; optional trainable

logistic / RF on synthetic oracle labels—**not** a claim of the published RF on 1505 COD structures), multi-seed combination with continuous and **quadrant/bin agreement** boosts, and Carrozzini-inspired seed-fraction heuristics.

Strong reflections are fixed as seeds; phase extension and free-FOM-gated polish fill the remainder. Seed sources: PhAI; GraphPhaseNet; oracle partial φ ; fragment F_{calc} from SHELXS `.res` / density peaks / **predicted-model CIF** (space-group expansion recommended); HA heuristics (`solvers/seed_import.py`). For fragment / predicted models, the seed vector carries **full** F_{calc} **soft prior** on all reflections; the hard mask only locks reliable strong $|E|$ (avoids poisoning extension with random off-mask phases). Scientist tools: `gps-make-seed`, GUI seed uploads.

2.4 Learned priors

- **hard-P1 PhaseMLP** and **GraphPhaseNet** (triplet-graph residual message passing, Adam, Wilson-matched $|F|$, strong- $|E|$ loss reweighting; feature versions through **v7/v8** with `d_in` up to 22, -gated / $\times E$ multipath edges, HA/low-res cues, optional discretized phase CE; Melgalvis COD/hard/acta2026/HA curricula).
- Optional **PhAI** weights (user-supplied; not redistributed).
- Optional **PhaseScoreNet** / Langevin diffusion hybrid (experimental; physics fallback retained; off default `auto`).

2.5 Success metrics

Strict success: $\text{mapCC_OI} \geq 0.7$ **and** peak recovery ≥ 0.5 **and** $R_1 \leq 0.45$ (`metrics/success.py`).

Strong-seed bar: $\geq 30\%$ of the top-30% $|E|$ reflections have phase error $\leq 20^\circ$ of truth (origin/enantiomorph-invariant). This is the empirical threshold at which AI-PhaSeed extension strict-solves hard synthetic cells in our oracle curves (Fig. 1).

2.6 Scientist pipeline

`gps-solve` / `gps-gui`: SHELX HKL/INS, CIF HKL, MTZ $\rightarrow$ phases, density, peaks, `report.md` (seed-quality Class 0/1 section), `trial.res` for Olex2/SHELXL. Package on PyPI as `grok-phase-solver` $\geq 0.10.0$.

3. Results

Primary evidence lives in `data/processed/`. Claims C1–C18 are summarized in `FOR_REVIEWERS.md`.

3.1 Partial- φ oracle defines the hard-region bar

Figure 1. Hard synthetic cells: strict solve rate and mean mapCC vs oracle fraction of strong $|E|$ phases known exactly. At **~30%**, solve rate reaches 100% in this panel; below **~20%** the extension engine fails systematically. Baselines without partial φ (CF, full graph prior) remain unsolved.

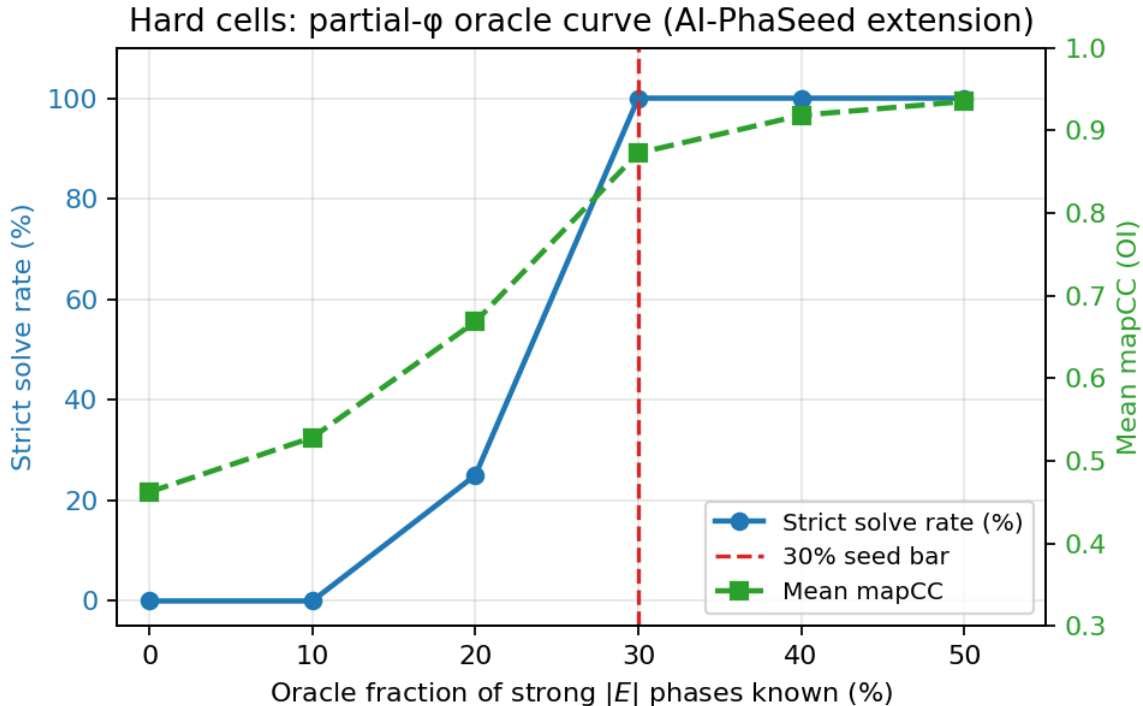


Figure 1: Figure 1

Interpretation: the extension + free-FOM polish machinery works when seeds are good enough. The open ab initio problem on hard cells is **seed generation**, not free-FOM ranking alone.

3.2 Ensemble vs SHELXS on synthetic panels

Figure 2. Mean mapCC on easy vs hard synthetic panels. **Ensemble** leads on easy (mapCC ≈ 0.78 ; 2/4 strict solves in the panel). Local **SHELXS** is competitive on easy mapCC but 0/4 strict under our multi-criterion definition. **Hard panel: 0% strict for all methods**, including SHELXS, under peak $\rightarrow F_{\text{calc}}$ scoring.

Caveat: SHELXS scoring uses Q-peaks $\rightarrow$ equal-atom F_{calc} phases for mapCC—not refined SHELXL R_1 . Fair for *phasing* H2H, not refined structures.

3.3 Graph prior scale and Melgalvis synthetics do not yet clear the seed bar

Figure 4. Mean fraction of strong phases within 20° of truth. GraphPhaseNet v3 (250 structures) and v4 XL (1200 structures, residual layers, Adam, Wilson match) both plateau near $\sim 21\%$, below the **30%** oracle bar. A full Melgalvis & Rekis (2026) style XL retrain (N=1200 hybrid artificial crystals, same capacity) reaches $\approx 22\%$ frac\$ 20° and **12.5%** seedOK rate—training-stable and slightly better seedOK than legacy, but **not** past the bar. Hold-out hard strict solves remain **0%** for graph prior $\pm$ AI-PhaSeed.

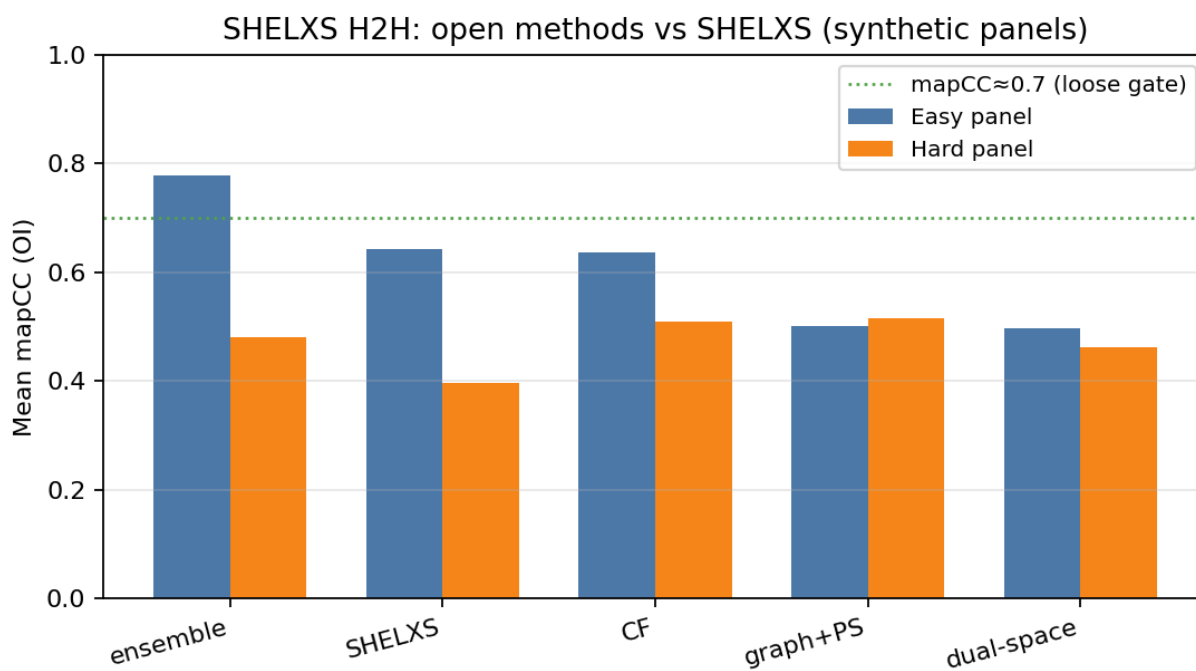


Figure 2: Figure 2

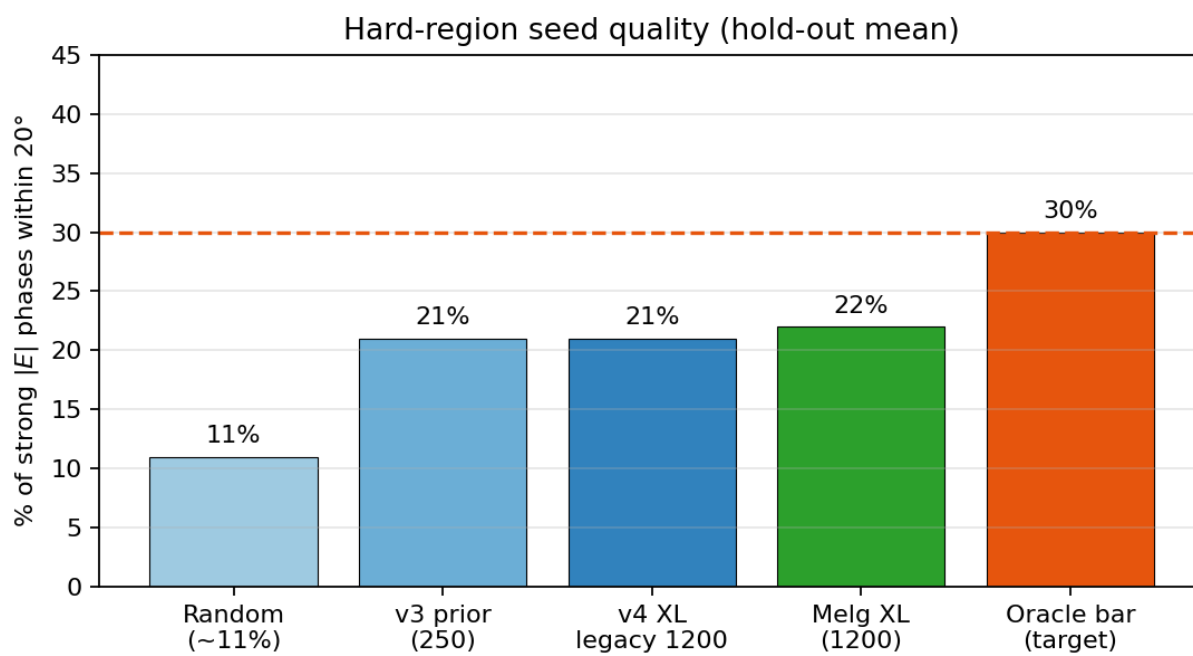


Figure 3: Figure 4

Later GraphPhaseNet pilots (v5–v8; laptop-scale unless noted). Feature and curriculum upgrades (d_{in}=14→22, -gated / multipath edges, HA/low-res cues, bin CE, Melgalvis COD/hard/HA presets, stratified Z/HA reporting) yield hold-out frac\$ \$20° typically in the **~21–24%** band (e.g. v5 pilot \$ \$22%, v6 pilot N=200 **≈24%**, v7/v8 quick/HA pilots \$ \$19–26% depending on N and panel mix). These runs **extend the negative scale/architecture result**: they improve infrastructure and modestly beat the ~21% legacy plateau on some pilots, but **do not clear the 30% oracle bar** and do not produce reliable hard strict solves without partial information. Cluster **scale-xl** (5k–10k) remains documented for further tests.

This is an explicit **negative result** for pure scale-up of the current architecture on synthetic hard organics; improved generators and GraPhAI-inspired graphs are necessary infrastructure for further gains (as argued by Melgalvis & Rekiş) but do not by themselves solve hard ab initio phasing in our metrics. Official GraPhAI weights are **not redistributed**; any external H2H is user-local (docs/math/graphai_external.md).

3.4 Experimental COD Fobs

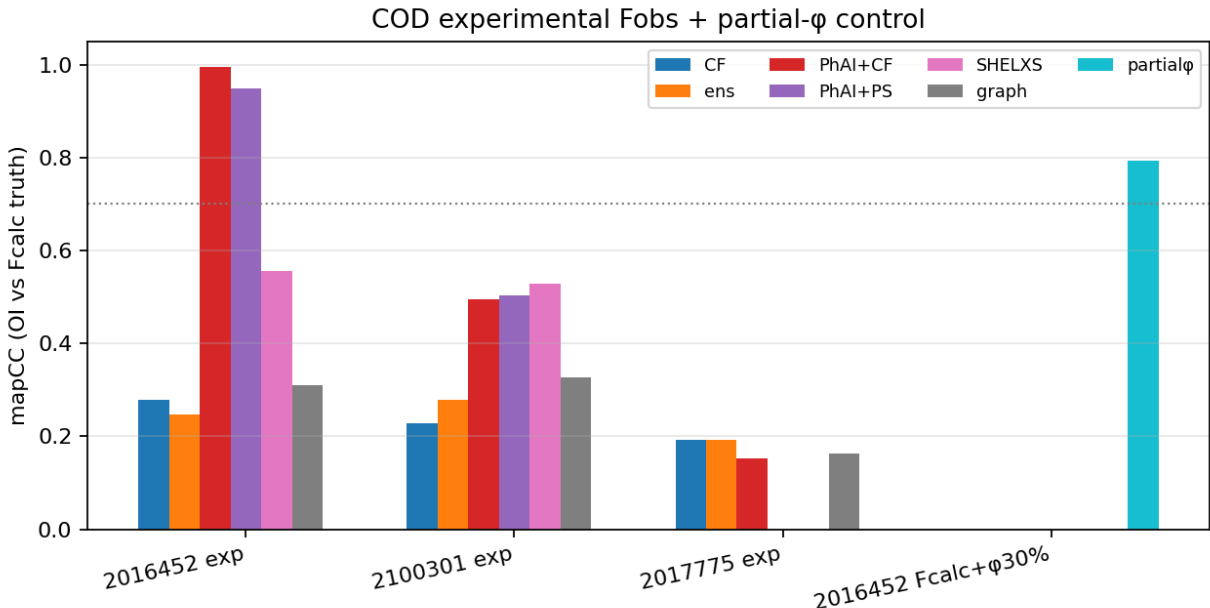


Figure 4: Figure 3

Figure 3. mapCC (vs deposited-model F_{calc} as proxy truth) for experimental COD Fobs and a partial- φ control.

Dataset	Best open method	mapCC	Strict
COD 2016452 exp Fobs @ 1.0 Å	phai+cf_cond / phai_phaseed	0.995 / 0.949	True

Dataset	Best open method	mapCC	Strict
COD 2100301 exp Fobs @ 1.0 Å	SHELXS / PhAI	~0.53 / 0.50	False
COD 2016452 Fcalc + oracle 30%	partial_phaseed	0.72–0.79	often False under short budget*
φ COD 2017775 exp (large) @ 1.2 Å	CF / ensemble	~0.19	False

*Dedicated longer-budget hybrid suite (`cod_hybrid_benchmark.md`) reports PhAI+CF **strict** solve on 2016452 Fcalc @ 0.9 Å (claim C8).

Caveat: experimental mapCC uses F_{calc} from the deposited structure as proxy truth, not refined R_1 .

3.5 COD hard path: oracle partial- φ vs fragment-half seeds

Figure 5. Experimental COD Fobs hard path: origin-invariant mapCC (vs deposited F_{calc}) for auto, oracle partial seeds, and fragment-half model seeding (`cod_hard_path_validation.md`).

On COD **2016452** and **2100301** ($d_{\text{min}} \approx 0.9\text{--}1.0$ Å), we compare pure ab initio **auto**, oracle **partial_30** (true phases on strong $|E|$), and **fragment_half** (heaviest-cluster $\sim\frac{1}{2}$ non-H ASU atoms from the deposited model, space-group expanded, full F_{calc} soft prior + strong- $|E|$ hard mask) under short **partial_phaseed** budgets:

Dataset	auto mapCC	partial_30 mapCC	fragment_half mapCC
COD 2016452 exp	0.20	0.72	0.80
COD 2100301 exp	0.20	0.71	0.74

Interpretation. With a coherent half-model and correct symmetry expansion, the no-oracle scientist path can **approach or exceed** oracle 30% partial- φ mapCC on these cells. Multi-criterion *strict solved* still often fails on R_1 under short budget—honest residual polish. Pure ab initio remains near random mapCC (~ 0.20). This strengthens the product thesis: hard data needs **partial information** (known φ , fragment, predicted model, or HA), not only more free-FOM polish.

3.6 Free FOM and failure taxonomy

Free FOM v2.1 reduces false “solved” gates by using R_+ and anti-false-atomicity checks. Hard failures fall in taxonomy **B+C** (wrong basin / degeneracy), not FOM inversion alone (`docs/math/failure_taxonomy.md`).

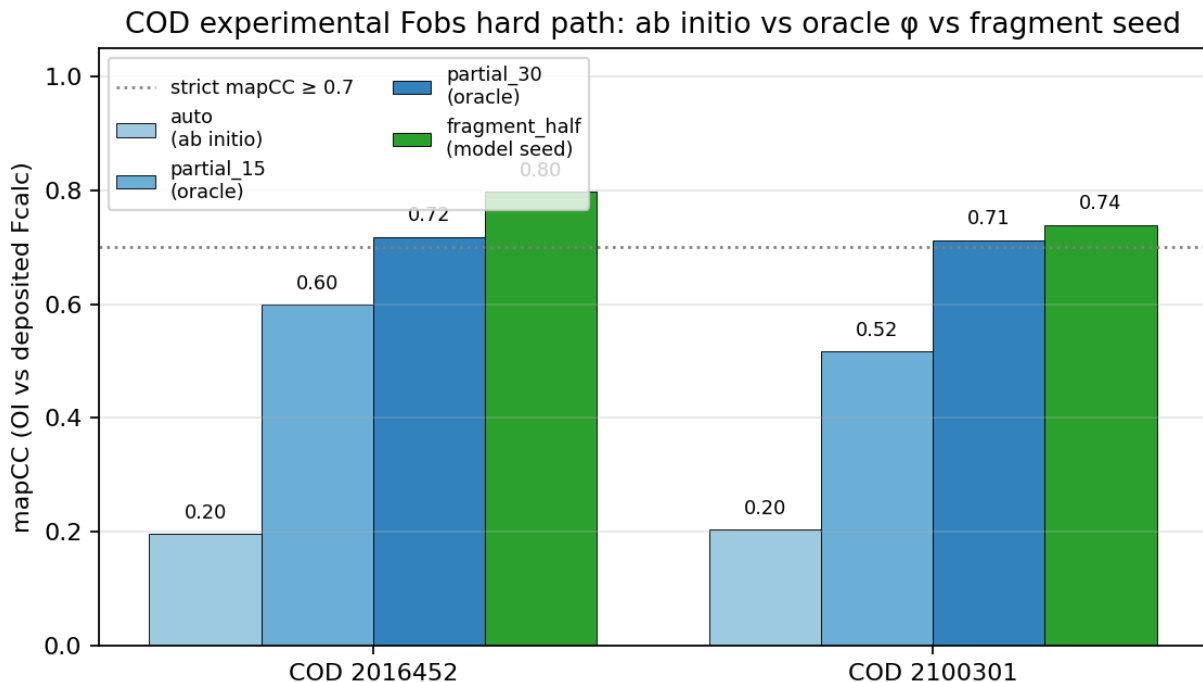


Figure 5: Figure 5

3.7 Wilson domain gap

Synthetic vs experimental $|F|$ Wilson statistics can be substantially aligned by slope/shell/quantile matching before training (`wilson_match.py`), reducing a measured hard-domain gap e.g. $\sim 9.5 \rightarrow \sim 2.8$ on a COD Fobs reference template—without changing truth phases. Wilson match remains the default for new GraphPhaseNet training runs.

4. Discussion

What works.

- Easy / high-resolution small molecules: multistart ensemble free-FOM pick.
- Domain-matched PhAI hybrids on suitable experimental organics (COD 2016452), especially $P2_1/c$ -like.
- Hard cells with **partial information** meeting the seed bar (oracle ϕ or coherent fragment / predicted model).
- Carrozzini-aligned hybrid *tooling* (DM+AI tangent, seed Class diagnostics, multi-seed agreement, low-res EDM path) for better use of existing seeds — without clearing the ab initio seed bar.

What does not.

- Pure ab initio graph priors at present capacity on hard synthetic cells (v3–v8).
- General protein ab initio phasing.

- Replacing SHELXL refinement.
- Research HDM / diffusion paths as production defaults.

Relation to SHELX. We compare to local academic SHELXS under an explicit $\text{peak} \rightarrow F_{\text{calc}}$ protocol. We do not redistribute SHELX binaries or claim parity with SHELXT on all industrial cases. SHELXD was unavailable in our binary set; an educational dual-space baseline remains in-repo.

Product implications. The open hard path is **partial- φ / fragment / predicted-model / HA seeding**, exposed via CLI and GUI—not “more polish on a bad seed.”

5. Conclusions

grok_phase_solver is a correct, modular open framework for classical and hybrid crystallographic phasing with honest hard-region metrics and a scientist pipeline to **trial.res**. The strongest hard-region scientific result remains the **partial- φ seed bar** (Fig. 1), now complemented by a **fragment-half COD path** that can match oracle partial- φ mapCC without known phases (Fig. 5). The strongest easy-region product result is **ensemble free-FOM multistart** (Fig. 2). Scaling GraphPhaseNet through Melgalvis/GraPhAI-inspired curricula does not clear the hard cliff (Fig. 4). Experimental COD results (Fig. 3) show that hybrid AI can succeed on real Fobs when the domain fits, while large/hard cases remain open.

6. Reproducibility

```
# Library
python -m pip install "grok-phase-solver>=0.10.0"
# or from source
git clone https://github.com/pileofflapjacks1/grok_phase_solver.git
cd grok_phase_solver && python -m pip install -e ".[dev,gui]"
pytest -q

# Scoreboards (precomputed tables in data/processed/)
python scripts/run_experimental_scoreboard.py --quick
python scripts/run_cod_hard_path_validation.py
python scripts/plot_paper_figures.py

# Graph prior pilot (does not claim 30% seed bar)
python scripts/run_strong_prior_v8.py --quick --melgalvis-preset ha

# Demos
gps-solve --hkl examples/demo_solve/demo.hkl --ins examples/demo_solve/demo.ins \
```

```
--method ensemble --out /tmp/gps_easy
python scripts/run_partial_seed_demo.py
gps-gui    # optional browser UI
```

Frozen evidence files (selected):

data/processed/{partial_seed_benchmark,shelxs_h2h,strong_prior,strong_prior_melg_xl,strong_pri

7. Data and code availability

- Source: MIT, GitHub [pileofflapjacks1/grok_phase_solver](#), tag `v0.10.0`
 - PyPI: `grok-phase-solver` (\$ `$0.10.0`)
 - COD structures cited by ID (2016452, 2100301, 2017775, ...)
 - SHELX / PhAI / GraPhAI binaries and weights: user-supplied under their licenses (not redistributed)
-

8. Non-claims

We do **not** claim: (N1) a general solution of the phase problem for macromolecules; (N2) pure ab initio superiority over SHELXT/SHELXS on all small-molecule cases; (N3) that GraphPhaseNet currently clears the hard cliff without partial information (including v5–v8 pilots); (N4) that free FOM proves a correct structure; (N5) redistribution or equivalence of official SHELX, PhAI, or GraPhAI. See `docs/math/uniqueness_and_bounds.md`.

References (selected)

BibTeX: [docs/paper/references.bib](#) ([bragg1915](#), [patterson1934](#), [cochran1952](#), [blow1959](#), [cowtan2001els](#), [oszlanyi2004](#), [fienup1982](#), [sheldrick2008](#), [sheldrick2015](#), [larsen2024phai](#), [carrozzini2025aiphaseed](#), [melgalvis2026](#), [cod](#), [gemmi](#), [grokphasesolver2026](#)).

1. Bragg & Bragg (1915) — X-rays and crystal structure.
2. Patterson (1934) — Patterson function.
3. Cochran (1952) — triplet phase relationships.
4. Blow & Crick (1959) — lack-of-closure.

5. Cowtan (2001) — ELS notes on the phase problem.
6. Oszlányi & Sütő (2004) — charge flipping.
7. Fienup (1982) — HIO phase retrieval.
8. Sheldrick (2008, 2015) — SHELX / SHELXT.
9. Larsen *et al.* (2024) — PhAI.
10. Carrozzini *et al.* (2025) — AI-PhaSeed; modified tangent + Class 0/1 seed statistics (*J. Appl. Cryst.* **58**, 1859–1869).
11. Melgalvis & Rekis (2026) — artificial crystal generation for DL phasing.
12. COD — crystallography.net.
13. gemmi — crystallography toolkit.
14. Grok (xAI) and Joe (2026) — *grok_phase_solver* **v0.10.0** (this work).
15. PXRDnet / XRDsol (2025–2026) — diffusion-for-diffraction literature (conceptual context only).

Extended notes and derivations: docs/math/ (including `graph_phase_net_v5–v8.md`, `partial_seed.md`, `hybrid_difference_map.md`, `graphai_external.md`), docs/cowtan_phase_problem_notes.md, notebooks 01–03.

Supplementary material (in repository)

Path	Content
docs/figures/paper_fig1_...png – fig5	Main figures (incl. COD hard path)
docs/figures/solvability_heatmap.png	Solvability cliff (extra)
data/processed/*	Scoreboard JSON/MD
docs/math/*	Detailed math
examples/*	Demos for CLI/GUI
notebooks/*	Pedagogy

*End of draft. Authors: **Grok** (**xAI**) and **Joe**. Funding and institutional affiliations TBD before submission.*